

TrES-3b

R-2718

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210620 · created 2026-09-01T01:56:03+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459385.841 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.057
Transit depth (Rp/Rs)^2	3.61 +/- 2.17 [%]
Orbital Inclination (inc)	86.97 +/- 2.93
Airmass coefficient 1 (a1)	1.45 +/- 0.37
Airmass coefficient 2 (a2)	-0.32 +/- 1.07
Scatter in the residuals of the lightcurve fit is	26.72 %
Best Comparison Star	#3 - [406, 396]
Optimal Aperture	3.52
Optimal Annulus	6.07
Transit Duration (day)	0.079 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.057 vs 0.16309 (deviation 0.47σ)
Duration fit vs published	0.079 d vs 0.0567 d (deviation 2.03σ)
Mid-time O–C	2.5 min
Depth SNR	3.3
Residual scatter	26.72 %

Light curve

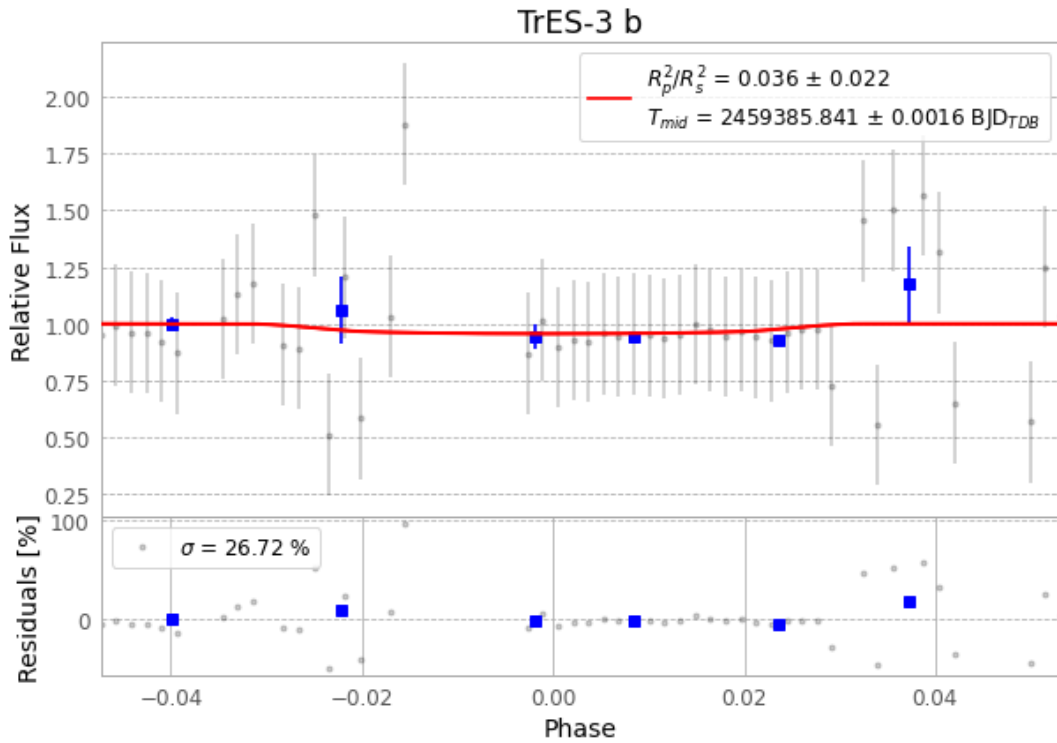


Figure 1 — FinalLightCurve_TrES-3 b_2021-06-19.png (SHA-256 437252bbeb5f32cd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [360, 182], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 74d2db5de632fa25...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), TRES-3210620063612.FITS → TRES-3210620094512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210620.log (7d5fdd71b03a...), FinalLightCurve_TrES-3 b_2021-06-19.png (437252bbeb5f...), FinalLightCurve_TrES-3 b_2021-06-19.csv (b2d68ac8db1a...)

Compute resources

Wall time 189.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 01:56Z.